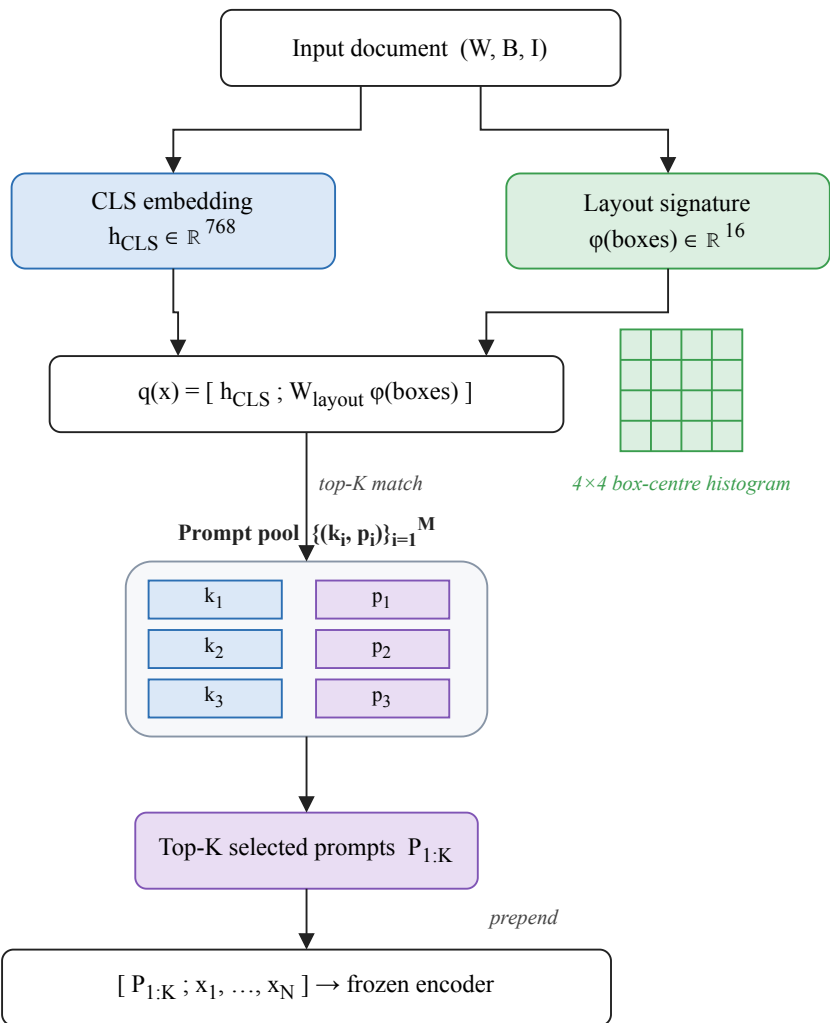
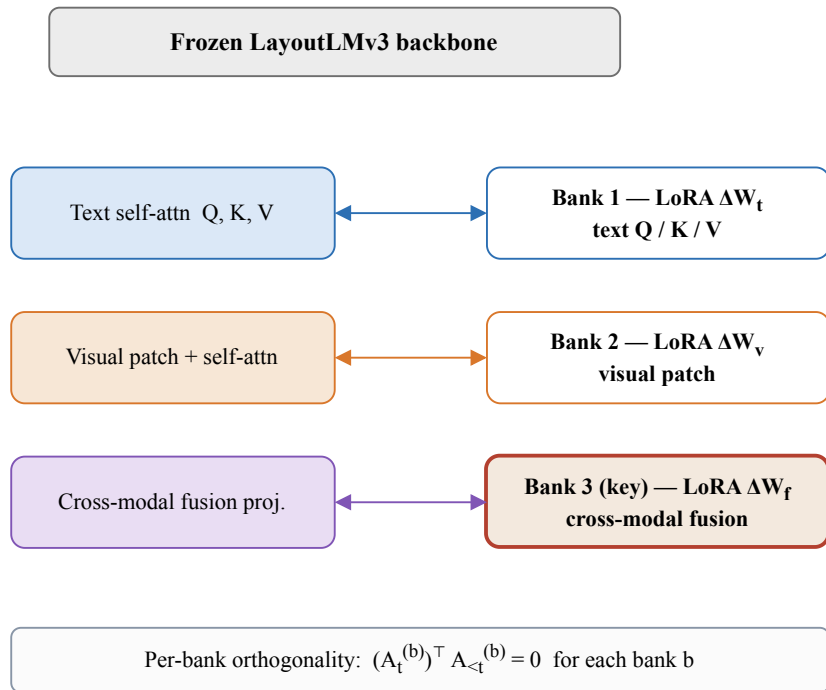


(A.1) Layout-Aware Prompt Pool (LAPP)



(A.2) Hierarchical LoRA (H-LoRA), per component



Triggered if fusion forgets most and layout-sensitivity matters; rehearsal-free (no stored exemplars).

■ Text stream (Bank 1)
 ■ Visual stream (Bank 2)
 ■ Layout / $\phi(\text{boxes})$

■ Fusion stream (Bank 3, key)

■ key bank (accent)